SET–20: STOCK ANALYSIS

Data Handling and Visualization – R Programming

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Experiment Title

Stock Analysis using R Programming

1. Aim

To analyze the prices of three stocks using R by creating a line chart, calculating the daily percentage change for Stock A and representing it using a bar chart, and displaying the stock-price data in tabular form.

2. Dataset

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

The SET-20 experiment specifies four activities: a line chart for the three stock prices, a bar chart for Stock A's daily percentage change, a table of stock prices, and a Tableau dashboard combining the visualizations.

3. Program 1 – Line Chart of Stock Prices

Aim: To create a line chart to visualize the stock prices of Stock A, Stock B, and Stock C over the given period.

```
# SET 20 - PROGRAM 1
# Line Chart of Stock Prices

# Create the dataset
date <- as.Date(c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
))

stock_a <- c(100, 105, 110)
stock_b <- c(150, 152, 148)
stock_c <- c(120, 118, 122)

stock_data <- data.frame(
  Date = date,
  Stock_A = stock_a,
  Stock_B = stock_b,
  Stock_C = stock_c
)

# Plot Stock A
plot(
  stock_data$Date,
  stock_data$Stock_A,
  type = "o",
  col = "blue",
  pch = 16,
  lwd = 2,
  ylim = range(
    c(stock_data$Stock_A,
      stock_data$Stock_B,
```

```
      stock_data$Stock_C)
  ),
  main = "Stock Price Trend",
  xlab = "Date",
  ylab = "Stock Price"
)

# Add Stock B
lines(
  stock_data$Date,
  stock_data$Stock_B,
  type = "o",
  col = "red",
  pch = 17,
  lwd = 2
)

# Add Stock C
lines(
  stock_data$Date,
  stock_data$Stock_C,
  type = "o",
  col = "green",
  pch = 15,
  lwd = 2
)

# Add legend
legend(
  "topright",
  legend = c("Stock A", "Stock B", "Stock C"),
  col = c("blue", "red", "green"),
  pch = c(16, 17, 15),
  lwd = 2
)
```

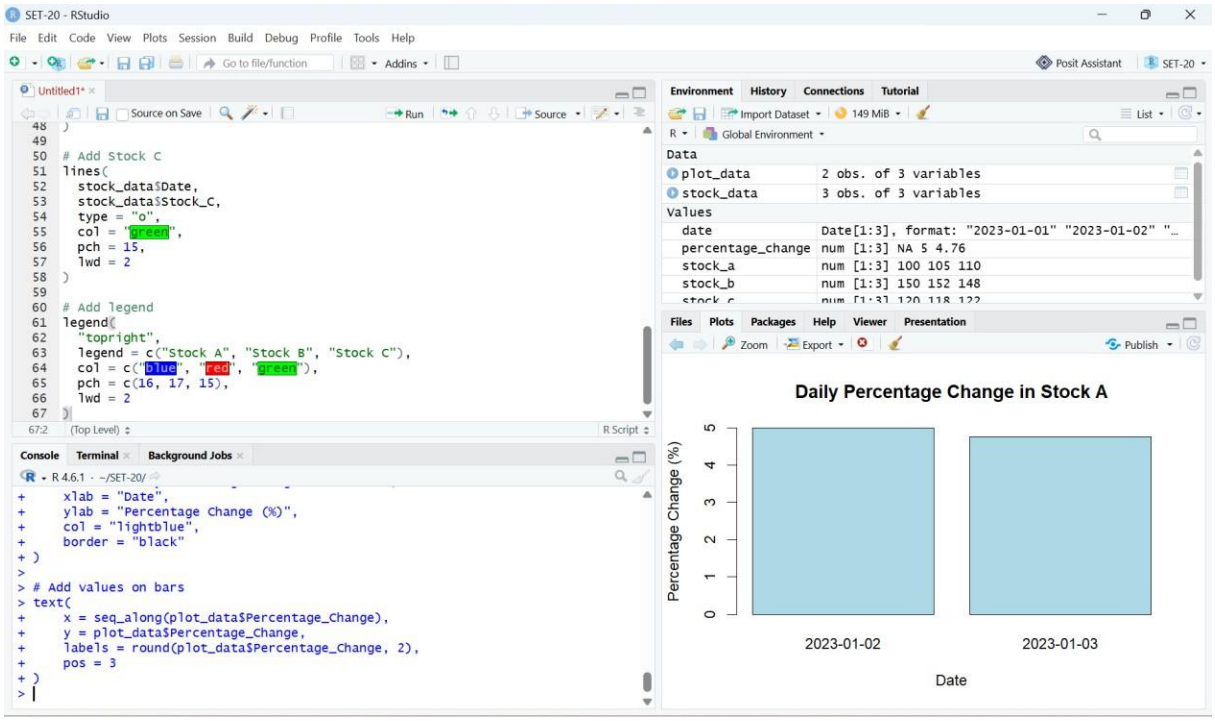


Figure 1: RStudio output – Stock Price Trend

Observation

- Stock A increases from 100 to 110.
- Stock B increases slightly from 150 to 152 and then decreases to 148.
- Stock C decreases from 120 to 118 and then increases to 122.
- Stock B has the highest price among the three stocks during the given period.

4. Program 2 – Daily Percentage Change in Stock A

Aim: To calculate the daily percentage change in Stock A and represent the result using a bar chart.

Formula:

Percentage Change = ((Current Price − Previous Price) / Previous Price) × 100

```
# SET 20 - PROGRAM 2
# Daily Percentage Change in Stock A

date <- as.Date(c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
))

stock_a <- c(100, 105, 110)

stock_data <- data.frame(
  Date = date,
  Stock_A = stock_a
)

# Calculate daily percentage change
percentage_change <- c(
  NA,
  diff(stock_data$Stock_A) /
    stock_data$Stock_A[-length(stock_data$Stock_A)] * 100
)

stock_data$Percentage_Change <- percentage_change

# Display calculated data
print(stock_data)

# Remove first NA value for plotting
plot_data <- stock_data[-1, ]

# Create bar chart
barplot(
  plot_data$Percentage_Change,
  names.arg = format(plot_data$Date, "%Y-%m-%d"),
  main = "Daily Percentage Change in Stock A",
  xlab = "Date",
  ylab = "Percentage Change (%)",
  col = "lightblue",
  border = "black"
)
```

```
# Add values on bars
text(
  x = seq_along(plot_data$Percentage_Change),
  y = plot_data$Percentage_Change,
  labels = round(plot_data$Percentage_Change, 2),
  pos = 3
)
```

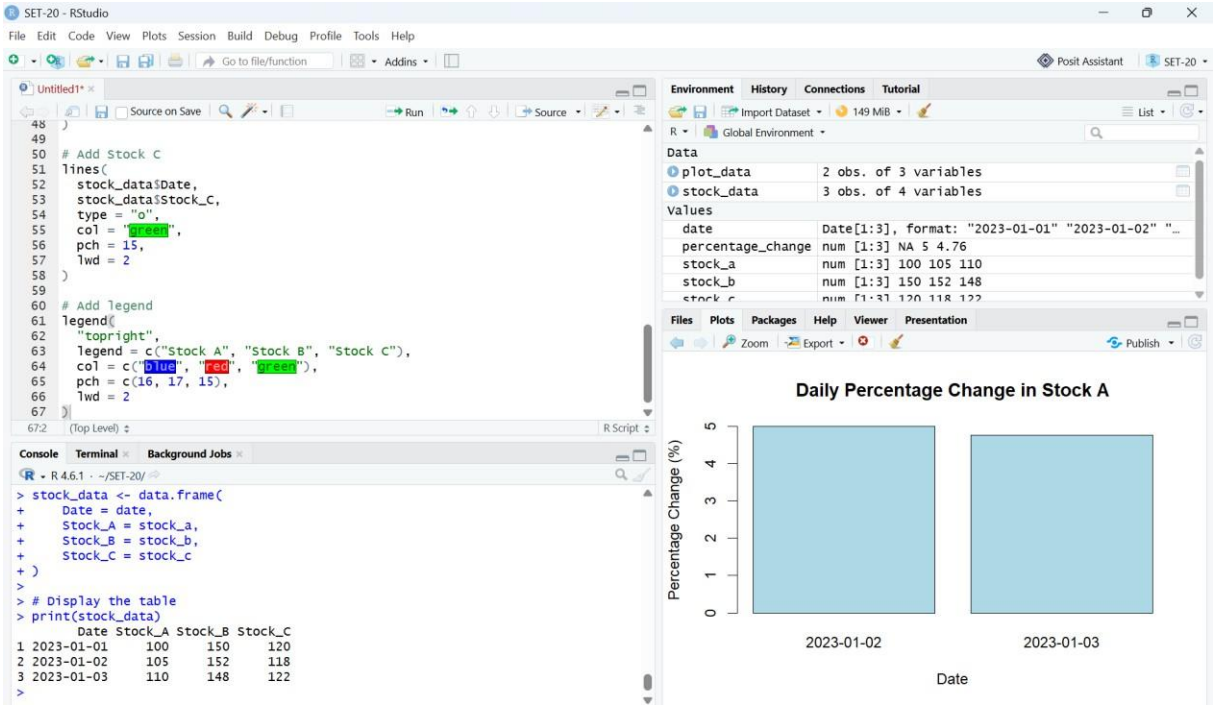


Figure 2: RStudio output – Daily Percentage Change in Stock A

Expected Result

Date	Stock A	Daily Percentage Change
2023-01-02	105	5.00%
2023-01-03	110	4.76%

Observation

Stock A shows a positive percentage change on both days. The increase is 5.00% on 2023-01-02 and approximately 4.76% on 2023-01-03, indicating continued upward movement over the period.

5. Program 3 – Stock Price Data Table

Aim: To build a table displaying the stock prices of all three companies for each date.

```
# SET 20 - PROGRAM 3
# Stock Price Data Table

date <- as.Date(c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
))
```

```
stock_a <- c(100, 105, 110)
stock_b <- c(150, 152, 148)
stock_c <- c(120, 118, 122)

stock_data <- data.frame(
  Date = date,
  Stock_A = stock_a,
  Stock_B = stock_b,
  Stock_C = stock_c
)

# Display the table
print(stock_data)
```

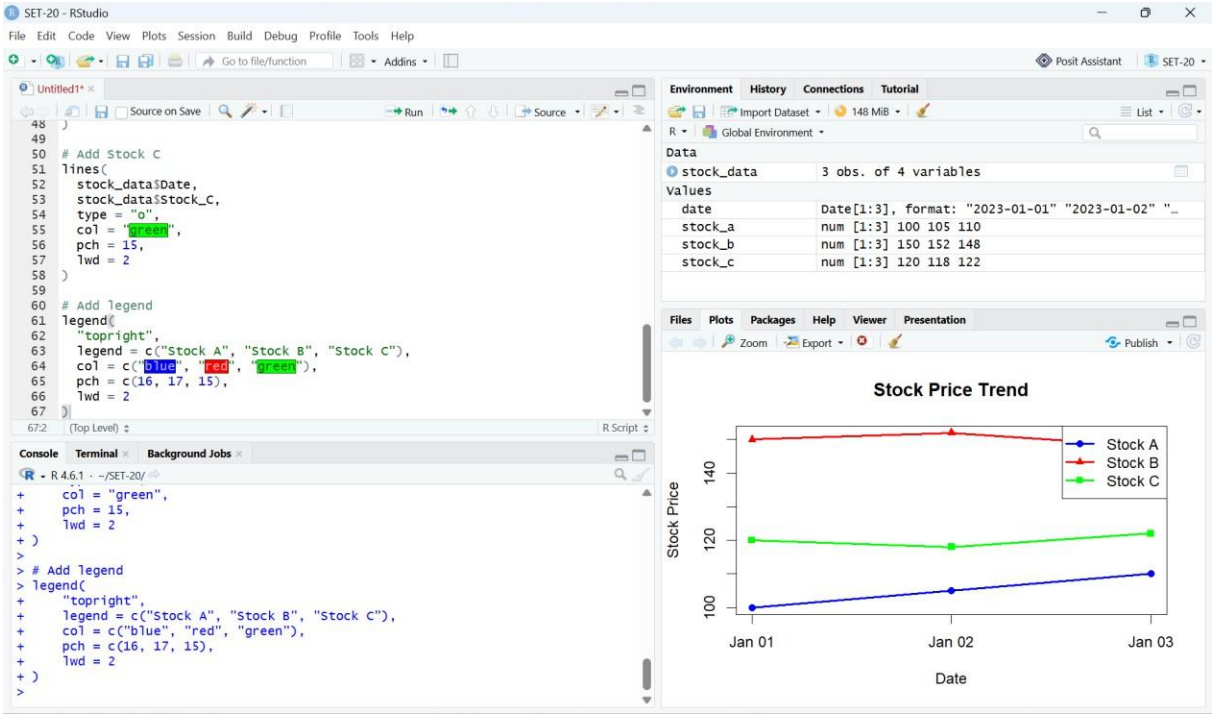


Figure 3: RStudio output – Stock Price Data / RStudio Environment

Expected Output

```
         Date Stock_A Stock_B Stock_C
1 2023-01-01      100      150      120
2 2023-01-02      105      152      118
3 2023-01-03      110      148      122
```

6. Program 4 – Tableau Dashboard

The fourth activity in SET-20 is to develop a Tableau dashboard combining the line chart, bar chart, and table for interactive exploration of stock prices.

Tableau Dashboard Components

- Line Chart: Date on Columns and Stock A, Stock B, and Stock C as the stock-price measures.
- Bar Chart: Date on Columns and Daily Percentage Change of Stock A on Rows.
- Table: Date, Stock A, Stock B, and Stock C displayed as a data table.
- Dashboard: Combine all three sheets into a single interactive dashboard.

Suggested Dashboard Layout

Top section: Stock Price Trend line chart
Bottom-left: Daily Percentage Change in Stock A bar chart
Bottom-right: Stock Price Data table

7. Overall Result

The Stock Analysis experiment was completed using R. A line chart was created to compare Stock A, Stock B, and Stock C, the daily percentage change of Stock A was calculated and visualized using a bar chart, and the stock-price data was displayed in tabular form. The three outputs can be combined with a Tableau dashboard for interactive exploration.

8. Conclusion

The experiment demonstrates how R can be used to organize stock-price data, visualize trends, calculate percentage changes, and present results clearly. The analysis shows that Stock A had continuous growth during the given three-day period, while Stocks B and C showed small fluctuations.